brittleness, strength, stiffness, machinability, malleability, hardness, toughness etc.

9.3.3 ELECTRICAL PROPERTIES

Electrical properties determine the ability of material to permit or resist the flow of electricity. Ex: conductivity, resistivity, di- electric, insulation, etc..

9.3.4 MAGNETIC PROPERTIES

Magnetic properties determine the behaviour of the materials with presence of applied magnetic field. Ex: permeability, retenativity, hysteresis, curie temperature, etc..

9.3.5 OPTICAL PROPERTIES

Optical properties of a material determine the behaviour of a material under the action of light. Ex: emission of light, absorbance, color luminosity, photo sensitivity, reflecting, refractive index, scattering, transmission etc.

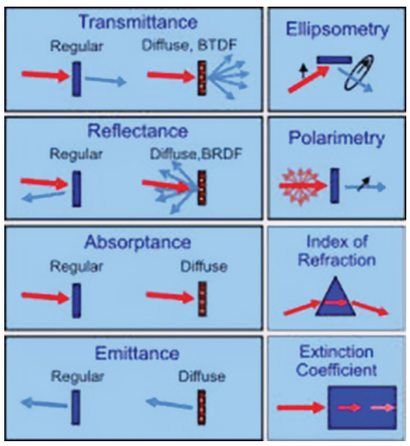


Fig 9.4 *Optical Properties*

9.3.6 CHEMICAL PROPERTIES

Chemical properties determine the corrosion rate, chemical reaction rate of material. Ex: corrosion resistance, reactivity, chemical composition, PH, hydroscopy etc.

9.3.7 THERMAL PROPERTIES

Thermal properties determine the behaviour of the materials when they are subjected to thermal changes. Ex: specific heat, thermal conductivity , thermal expansion, thermal stress, thermal shock, latent heat, specific heat , etc..

9.4 MECHANICAL PROPERTIES

Mechanical property defines the behaviour of material under an act of force or load.

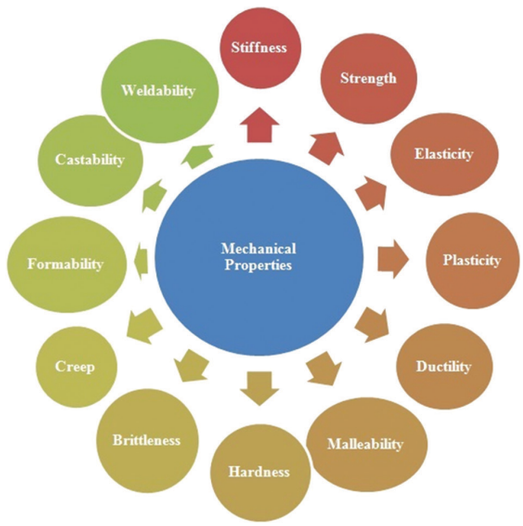


Fig 9.5 *Mechanical Properties*

i) Strength

It is defined as the ability of a material to resist loads without failure and fracture.

ii) Stiffness

It is defined as the ability of a material to resist deformation or deflection under load. Stiffness within the elastic limit is known as modulus of Elasticity.

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